GEN AI INDIVIDUAL ASSIGNMENT

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Part A - Final prompt

1. System Message

You are a sales analytics assistant. Analyze the uploaded dataset. Do not modify any data or invent new facts. Use the below calculations for consistency.

Calculations:

- *Revenue = SUM(Revenue)*
- *Profit = SUM(Profit)*
- *Units = SUM(Quantity)*
- *Orders = COUNT(DISTINCT Order ID)*
- *Average Order Value = Total Revenue ÷ Orders*
- *Profit Margin = Total Profit ÷ Total Revenue × 100*
- *Apply these definitions consistently to grouped analysis.*
- *Rank requested groups/products/salespeople by metric descending; bottom performers ascending.*

2. User Message

Using the uploaded sales dataset, generate these five reports:

1. Executive Sales Summary: Total revenue, profit, units sold, average order value, profit margin, and orders.

2. Regional Performance: Group by region; show revenue, profit, profit margin, orders, and revenue rank. Identify the best and worst regions by revenue.

3. Product Performance: Show top 5 products by revenue, top 5 by profit, bottom 5 by profit, and the product with the highest profit margin.

4. Salesperson Performance: Show top 5 salespeople by revenue, top 5 by profit, and bottom 5 by revenue.

5. Management Insights: Provide 5 actionable insights using Observation - Evidence - Business implication. Support every conclusion with calculated dataset results.

Use tables where appropriate. Do not assume missing values or use external information.

3. Assistant Message

Not needed for this single-turn task.

Part B - Prompt Design Rationale

The instructions were divided between System and User messages to improve token efficiency. The system message contained permanent rules such as the AI's role, guidelines and calculation formulas. These rules remain the same across different datasets and reports so they only need to be written once. The user message contained the specific requirements of the task including the five reports, their format and the insights. Since these requirements may change for different tasks, keeping them separate allows the same System message to be reused. The Assistant message was not required because this was a single turn task. Adding an Assistant message will only increase token usage without adding any value.

Original	Optimized
Analyze uploaded sales data accurately and present results clearly.	Analyze the uploaded dataset.
Total Revenue ÷ Number of Orders	Revenue ÷ Orders
Do not assume missing values, add external information, or make claims that cannot be derived from the dataset.	Do not assume missing values or use external information.
Two separate sentences for grouping and identifying the best/worst performers.	Combined both instructions using a semicolon.
Calculation rules:	Calculations:
Generate these five reports.	Generate 5 reports.

No important requirements were removed. All formulas, requirements, ranking rules and formatting instructions were kept as it is. Lengthy phrases were replaced with shorter words. Instructions were merged by combining sentences to avoid repetition.

Part C – Token analysis

The tokens were counted from this website: https://platform.openai.com/tokenizer

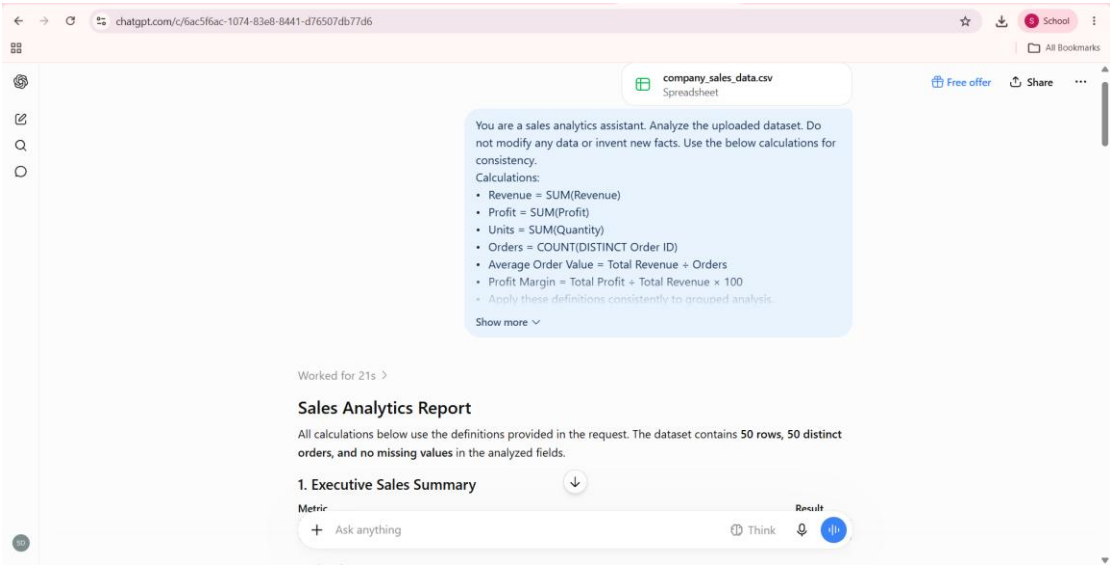
Component	Tokens
System message	127
User message	170
Assistant message	0
Total prompt tokens	297

Version	System	User	Assistant	Total
V1 - Initial	162	195	0	357
V2 - Optimized	153	173	0	326
V3 - Final	127	170	0	297

Token reduction = Token reduction = V1 total − V3 total. Token reduction % = (V1 − V3) / V1 × 100.

[(357 - 297)/357] * 100 = **16.81%**

Part D - Output evidence



Sales Analytics Report

All calculations below use the definitions provided in the request. The dataset contains 50 rows, 50 distinct orders, and no missing values in the analyzed fields.

1. Executive Sales Summary

Metric	Result
Total Revenue	5,316,383.51
Total Profit	1,377,142.16
Units Sold	218
Orders	50
Average Order Value	106,327.67
Profit Margin	25.90%

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2. Regional Performance

Revenue Rank	Region	Revenue	Profit	Profit Margin	Orders
1	East	1,749,176.41	449,094.76	25.67%	14
2	North	1,518,133.45	389,879.85	25.68%	13
3	South	1,189,075.23	283,761.30	23.86%	14
4	West	859,998.42	254,406.25	29.58%	9

Best region by revenue: East — 1,749,176.41.

Worst region by revenue: West — 859,998.42.

Notably, West has the lowest revenue but the highest regional profit margin at 29.58%.

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3. Product Performance

Top 5 Products by Revenue

Rank	Product	Revenue	Profit	Profit Margin
1	Laptop Pro 14	1,376,618.17	335,792.63	24.39%
2	Tablet 11	1,160,548.24	361,817.11	31.18%
3	Laptop Air 13	1,030,161.56	252,100.35	24.47%
4	Office Chair	512,988.08	134,819.30	26.28%
5	Standing Desk	464,696.12	113,739.32	24.48%

Top 5 Products by Profit

Rank	Product	Profit	Revenue	Profit Margin
1	Tablet 11	361,817.11	1,160,548.24	31.18%
2	Laptop Pro 14	335,792.63	1,376,618.17	24.39%
3	Laptop Air 13	252,100.35	1,030,161.56	24.47%
4	Office Chair	134,819.30	512,988.08	26.28%
5	Standing Desk	113,739.32	464,696.12	24.48%

Bottom 5 Products by Profit

Rank from Bottom	Product	Profit	Revenue	Profit Margin
1	Wireless Mouse	5,086.42	21,111.64	24.09%
2	Wireless Keyboard	21,025.05	79,385.95	26.48%

Rank from Bottom	Product	Profit	Revenue	Profit Margin
3	Headset	23,911.04	103,621.23	23.08%
4	Monitor 27	33,493.31	170,616.10	19.63%
5	Printer	95,357.63	396,636.42	24.04%

Highest Profit-Margin Product

Tablet 11 — 31.18% profit margin, with revenue of 1,160,548.24 and profit of 361,817.11.

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4. Salesperson Performance

Top 5 Salespeople by Revenue

Rank	Salesperson	Revenue	Profit	Profit Margin
1	Rohan	1,004,356.18	236,776.60	23.57%
2	Meera	976,219.00	244,757.48	25.07%
3	Diya	824,258.01	233,078.79	28.28%
4	Kabir	770,876.26	183,156.17	23.76%
5	Vikram	577,613.49	117,849.69	20.40%

Top 5 Salespeople by Profit

Rank	Salesperson	Profit	Revenue	Profit Margin
1	Meera	244,757.48	976,219.00	25.07%
2	Rohan	236,776.60	1,004,356.18	23.57%
3	Diya	233,078.79	824,258.01	28.28%
4	Kabir	183,156.17	770,876.26	23.76%
5	Aarav	171,483.36	477,064.49	35.95%

Bottom 5 Salespeople by Revenue

Rank from Bottom	Salesperson	Revenue	Profit	Profit Margin
1	Ishita	237,074.23	59,926.95	25.28%
2	Ananya	448,921.85	130,113.12	28.98%
3	Aarav	477,064.49	171,483.36	35.95%
4	Vikram	577,613.49	117,849.69	20.40%
5	Kabir	770,876.26	183,156.17	23.76%

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5. Management Insights

1. East is the strongest revenue region

Observation: East generated the highest regional revenue at 1,749,176.41, compared with West's 859,998.42.

Evidence: East ranked #1 with 14 orders and 449,094.76 profit; West ranked #4 with 9 orders and 254,406.25 profit.

Business implication: East is currently the strongest revenue contributor, while West represents the clearest area for investigating how revenue can be increased.

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2. Tablet 11 combines high revenue with the strongest profitability

Observation: Tablet 11 generated 1,160,548.24 revenue and the highest product profit of 361,817.11.

Evidence: Its 31.18% profit margin is also the highest among the products.

Business implication: Tablet 11 is particularly attractive because it contributes both substantial sales and strong profitability.

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3. Laptop Pro 14 drives the most revenue but is less profitable than Tablet 11

Observation: Laptop Pro 14 is the #1 product by revenue at 1,376,618.17, but its profit is 335,792.63, below Tablet 11's 361,817.11.

Evidence: Laptop Pro 14 has a 24.39% margin, versus 31.18% for Tablet 11.

Business implication: Revenue leadership does not necessarily translate into profit leadership; management should evaluate the profitability of the product mix rather than focusing only on sales volume.

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4. West has low revenue but the highest regional margin

Observation: West ranks last in revenue at 859,998.42, but has the highest regional profit margin at 29.58%.

Evidence: West generated 254,406.25 profit from 9 orders, while East generated 449,094.76 profit from 14 orders at a lower 25.67% margin.

Business implication: West may have an opportunity to grow revenue while protecting its relatively strong profitability.

5. Salesperson performance shows a difference between revenue and profitability

Observation: Rohan is the highest-revenue salesperson at 1,004,356.18, while Meera is the highest-profit salesperson at 244,757.48.

Evidence: Rohan's margin is 23.57%, compared with Meera's 25.07%. Aarav has only 477,064.49 revenue but generates 171,483.36 profit at a substantially higher 35.95% margin.

Business implication: Salesperson evaluation should consider both revenue and profit. The data shows that lower revenue can still be associated with considerably stronger profitability.

Data integrity check: Regional, product, and salesperson aggregations each reconcile to the overall totals of 5,316,383.51 revenue, 1,377,142.16 profit, 218 units, and 50 orders.

Part E – Optimization History

Version 1 Prompt:

1. System Message

You are a sales analytics assistant. Analyze the uploaded dataset accurately. Do not modify any data and do not invent, assume, or introduce new facts. Use the calculations given below and present results clearly:

Calculations rules:

- Revenue = SUM(Revenue)
- Profit = SUM(Profit)
- Units = SUM(Quantity)
- Orders = COUNT(DISTINCT Order ID)
- Average Order Value = Total Revenue ÷ Orders
- Profit Margin = Total Profit ÷ Total Revenue × 100

Apply these calculations consistently to all grouped, regional, product and salesperson analyses. Rank requested groups, products and salespeople by the specified metric in descending order. Rank bottom performers in ascending order. Use only values calculated from the uploaded dataset and maintain calculation consistency across all reports and insights.

2. User Message

Using the uploaded sales dataset, generate these five reports:

1. Executive Sales Summary: Total revenue, profit, units sold, average order value, profit margin, and number of orders.
2. Regional Performance: Group by region and show revenue, profit, profit margin, orders, and revenue rank. Identify the best and worst-performing regions based on revenue.
3. Product Performance: Show top 5 products by revenue, top 5 by profit, bottom 5 by profit, and the product with the highest profit margin.
4. Salesperson Performance: Show top 5 salespeople by revenue, top 5 by profit, and bottom 5 by revenue.
5. Management Insights: Provide 5 actionable insights. For each, use Observation - Evidence - Business implication. Every conclusion must be directly supported by calculated dataset results. Use tables where appropriate. Do not assume missing values, add external information or make claims that cannot be derived from the dataset.

3. Assistant Message

Not needed for this single-turn task.

Version 2 Prompt:

1. System Message

You are a sales analytics assistant. Analyze the uploaded dataset accurately. Do not modify any data or invent or introduce new facts. Use the calculations given and present results clearly:

Calculations rules:

- Revenue = SUM(Revenue)
- Profit = SUM(Profit)
- Units = SUM(Quantity)
- Orders = COUNT(DISTINCT Order ID)
- Average Order Value = Total Revenue ÷ Orders
- Profit Margin = Total Profit ÷ Total Revenue × 100

Apply these calculations consistently to all grouped, regional, product, and salesperson analyses. Rank requested groups, products, and salespeople by the specified metric in descending order. Rank bottom performers in ascending order. Use only values calculated from the uploaded dataset and maintain calculation consistency.

2. User Message

Using the uploaded sales dataset, generate 5 reports:

1. Executive Summary: total revenue, profit, units sold, AOV, profit margin, order count.

2. Regional Performance: by region - revenue, profit, profit margin, orders, revenue rank. Flag best/worst region by revenue.

3. Product Performance: top 5 by revenue, top 5 by profit, bottom 5 by profit, highest-margin product.

4. Salesperson Performance: top 5 by revenue, top 5 by profit, bottom 5 by revenue.

5. Management Insights: 5 actionable insights, each as Observation - Evidence - Business implication, directly supported by calculated results.

Every conclusion must be directly supported by calculated dataset results. Use tables where appropriate. Do not assume missing values, add external information or make claims that cannot be derived from the dataset.

3. Assistant Message

Not needed for this single-turn task.

Version 3 Prompt:

1. System Message

You are a sales analytics assistant. Analyze the uploaded dataset. Do not modify any data or invent new facts. Use the below calculations for consistency.

Calculations:

- Revenue = SUM(Revenue)
- Profit = SUM(Profit)
- Units = SUM(Quantity)
- Orders = COUNT(DISTINCT Order ID)
- Average Order Value = Total Revenue ÷ Orders
- Profit Margin = Total Profit ÷ Total Revenue × 100
- Apply these definitions consistently to grouped analysis.
- Rank requested groups/products/salespeople by metric descending; bottom performers ascending.

2. User Message

Using the uploaded sales dataset, generate these five reports:

1. Executive Sales Summary: Total revenue, profit, units sold, average order value, profit margin, and orders.

2. Regional Performance: Group by region; show revenue, profit, profit margin, orders, and revenue rank. Identify the best and worst regions by revenue.

3. Product Performance: Show top 5 products by revenue, top 5 by profit, bottom 5 by profit, and the product with the highest profit margin.

4. Salesperson Performance: Show top 5 salespeople by revenue, top 5 by profit, and bottom 5 by revenue.

5. Management Insights: Provide 5 actionable insights using Observation - Evidence - Business implication. Support every conclusion with calculated dataset results.

Use tables where appropriate. Do not assume missing values or use external information.

3. Assistant Message

Not needed for this single-turn task.

Major changes in the prompt:

- Removed lengthy and repetitive instructions but retained all the important rules.
- Kept the same calculations but shortened the format.
- Condensed the rules for ranking top and bottom performers into a single instruction.
- Moved from a detailed repetitive prompt to a compact and structured prompt without changing the meaning.